# Chapter 14, Problem 47P

## Problem

Show that a series *LR* circuit is a low-pass filter if the output is taken across the resistor. Calculate the corner frequency *fc* if $L = 2$ mH and $R = 10$ kΩ.

## Step-by-step solution

**Step 1** of 8

Consider the following circuit diagram:

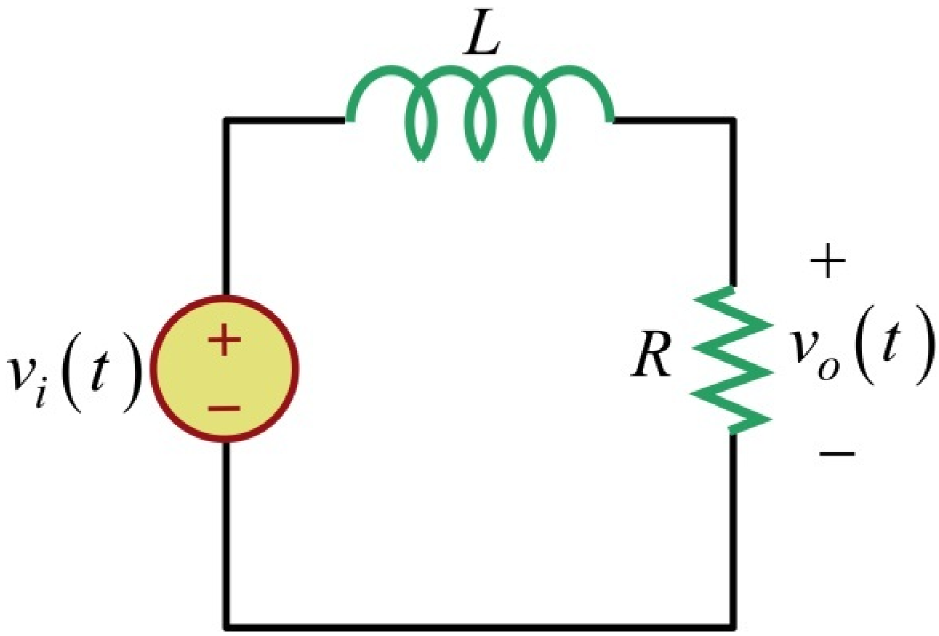


Figure 1

**Step 2** of 8

Calculate the transfer function $\mathbf{H}(\omega)$.

Write the expression for the transfer function $\mathbf{H}(\omega)$.

$$\mathbf{H}(\omega)=\frac{\mathbf{V}_o}{\mathbf{V}_i} \quad \ldots\ldots (1)$$

Draw the following circuit diagram to find the output voltage $\mathbf{V}_o$.

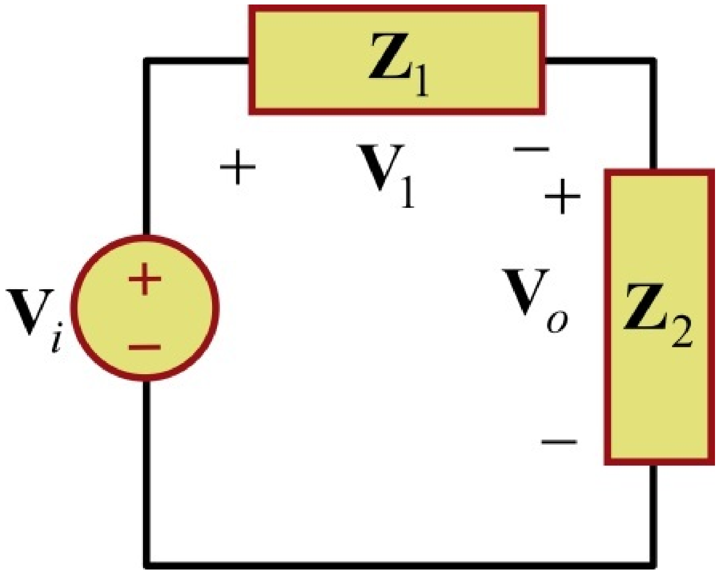


Figure 2

**Step 3** of 8

Calculate the output voltage $\mathbf{V}_o$.

Apply voltage division rule across the impedance $\mathbf{Z}_2$ to find the output voltage $\mathbf{V}_o$.

$$\mathbf{V}_o=\frac{\mathbf{Z}_2}{\mathbf{Z}_1+\mathbf{Z}_2}\mathbf{V}_i$$

Substitute $R$ for $\mathbf{Z}_2$, $j\omega L$ for $\mathbf{Z}_1$ in this expression.

$$\frac{\mathbf{V}_o}{\mathbf{V}_i}=\frac{R}{j\omega L+R}$$

Substitute equation 1 in this expression.

$$\mathbf{H}(\omega)=\frac{1}{1+j\omega\left(\frac{L}{R}\right)} \quad \ldots\ldots (1)$$

Therefore, the transfer function $\mathbf{H}(\omega)$ is $\frac{1}{1+j\omega\left(\frac{L}{R}\right)}$.

**Step 4** of 8

Calculate the value of $\mathbf{H}(0)$.

Substitute 0 for $\omega$ in the equation 1.

$$\begin{aligned}\mathbf{H}(0)&=\frac{1}{1+j(0)\left(\frac{L}{R}\right)}\\&=1\end{aligned}$$

Therefore, the value of $\mathbf{H}(0)$ is $1$.

Calculate the value of $\mathbf{H}(\infty)$.

Substitute $\infty$ for $\omega$ in the equation 1.

$$\mathbf{H}(\infty) = \frac{1}{1 + j(\infty)\left(\frac{L}{R}\right)}$$
$$= \frac{1}{\infty}$$
$$= 0$$

Therefore, the value of $\mathbf{H}(\infty)$ is $0$.



The values $\mathbf{H}(0)$ and $\mathbf{H}(\infty)$ are $1,\ 0$ respectively, which corresponds to the ideal characteristics of a Lowpass filter.

Hence, the $\boxed{\text{series } LR \text{ circuit}}$ is a low pass filter when the output is taken across the resistor as shown in Figure 1.

Draw the following circuit diagram:

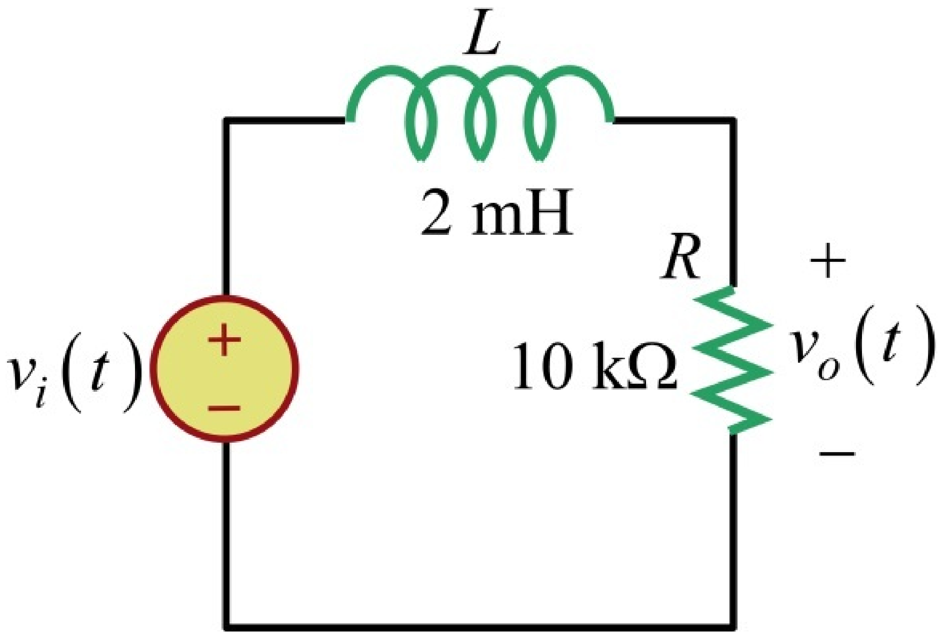


Figure 3

Calculate the magnitude of the transfer function $\mathbf{H}(\omega)$.

Write the expression for the magnitude of the transfer function $\mathbf{H}(\omega)$.

$$|\mathbf{H}(\omega)| = \left|\frac{1}{1+j\omega\left(\frac{L}{R}\right)}\right|$$

$$= \frac{1}{\left|1+j\omega\left(\frac{L}{R}\right)\right|}$$

$$|\mathbf{H}(\omega)| = \frac{1}{\sqrt{1+\omega^2\left(\frac{L}{R}\right)^2}} \quad \text{...... (2)}$$

Therefore, the magnitude of transfer function $\mathbf{H}(\omega)$ is $\frac{1}{\sqrt{1+\omega^2\left(\frac{L}{R}\right)^2}}$.

**Step 8** of 8

Calculate the value of the corner frequency $f_c$ of the circuit in Figure 3.

The corner frequency is the frequency at which the magnitude of transfer function $\mathbf{H}(\omega)$is $\frac{1}{\sqrt{2}}$

.

Substitute $\frac{1}{\sqrt{2}}$ for $|\mathbf{H}(\omega)|$ in the equation 2.

$$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{1+\omega_c^2\left(\frac{L}{R}\right)^2}}$$

$$1+\omega_c^2\left(\frac{L}{R}\right)^2 = 2$$

$$\omega_c^2\left(\frac{L}{R}\right)^2 = 1$$

$$\omega_c = \frac{R}{L}$$

Substitute $2\pi f_c$ for $\omega_c$ in this expression.

$$(2\pi f_c) = \frac{R}{L}$$

$$f_c = \frac{R}{L(2\pi)}$$

Substitute $2\text{ mH}$ for $L$ , $10\text{ k}\Omega$ for $R$ in this expression.

$$f_c = \frac{10\times10^3}{2\times10^{-3}(2\pi)}$$

$$= 0.7954\times10^6$$

$$\cong 796\times10^3$$

Therefore, the value of corner frequency $f_c$ is $\boxed{796\text{ kHz}}$.